RECALLING Article 28(b) of the Convention on the International Maritime Organization con cerning the functions of the Committee RECALLING ALSO its adoption, by resolution A.864(20), of the Recommendations for entering enclosed spaces aboard ships , and by resolutio n A.1050(27), of the Revised recommendations for entering enclosed spaces aboard ships , incorporating therein recommendations for entering cargo spaces, tanks, pump-rooms, fuel tanks, cofferdams, duct keels, ballast tanks and similar enclosed spaces, RECALLING FURT HER that SOLAS regulation III/19.3.6 requires enclosed space entry and --- URL: https:// wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MS C.581(110).pdf keels, ballast tanks and similar enclosed spaces, RECALLING FURTHER that SOLAS regulation III/19.3.6 requires enclosed space entry and rescue drills, SOLAS regul ation XI-1/7 requires the carrying of atmosphere testing instrument for enclosed spaces, and SOLAS chapter IX and the ISM Code require management for the safe operation of ships , RECALLING that the Maritime Safety Committee is requested to keep the revised recommen dations under review and amend them as necessary, BEING CONCERNED about the continued lo ss of life resulting from personnel entering shipboard spaces in which the atmosphere is unable to support life, owing to oxygen depletion, oxygen enrichment, toxicity or flamma bility, BEING AWARE of the work undertaken in this regard by the International Labour Or ganization, Governments and segments of the private sector, RECOGNIZING that organizatio nal leadership plays a crucial role in successful --- URL: https://wwwcdn.imo.org/localr esources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf Govern ments and segments of the private sector, RECOGNIZING that organizational leadership pla ys a crucial role in successful implementation of this guidance by empowering shipboard staff to make the right decisions, HAVING CONSIDERED , at its 110th session, the recomme ndation made by the Sub-Committee on Carriage of Cargoes and Containers at its tenth ses sion, 1 ADOPTS the Revised recommendations for entering enclosed spaces aboard ships, as set out in the annex to the present resolution; 2 INVITES Governments to bring the annex ed Revised Recommendations to the attention of shipowners, ship operators, seafarers, po rt and terminal operators and port workers, urging them to apply them, as appropriate, t o all ships and operations; 3 INVITES the Assembly to revoke resolution A.1050(27) and t o endorse the action taken by the Maritime Safety Committee.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E NCLOSED SPACES ABOARD SHIPS --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeC entre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf RESOLUTION MSC.581(110) (ado pted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING ENCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 2 I:\MSC\110\MSC 110-21-Add.3.docx ANNEX REVISED RECOMME NDATIONS FOR ENTERING ENCLOSED SPACES ABOARD SHIPS Preamble The objective of these recom mendations is to encourage the adoption of safety procedures and the development of soun d practices aimed at preventing casualties to, and enhancing the safety of, personnel en tering or working in enclosed spaces where there may be an oxygen-deficient, oxygen-enri ched, flammable and/or toxic atmosphere.

Additionally, in the event of an emergency within an enclosed space, these recommendatio ns provide guidance for an appropriate, planned and considered response complying with s hip-specific enclosed space rescue plans.

Investigations into the circumstances of enclosed space accidents have shown that a fail ure to --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResol utions/MSCResolutions/MSC.581(110).pdf Investigations into the circumstances of enclosed space accidents have shown that a failure to systematically identify the hazards, assess the risks and implement an appropriate entry procedure remains a significant factor in m any accidents.

Likewise, the complex structure of some spaces on board ships creates problems with appr opriate ventilation, illumination and movement within the space.

Organizational leadership on ship and shore plays a crucial role in successful implement ation of this guidance by empowering shipboard staff to make the right decisions.

These recommendations apply to all types of ships and provide guidance to ship operators , seafarers and shore personnel to work on board ships.

It should be noted that on ships where entry into enclosed spaces may be infrequent, for example, on certain pas senger ships or small general cargo ships, the dangers may be le ss apparent, and accordingly there may be a need for increased vigilance.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.581(110).pdf small general cargo ships, the dangers may be less appar

ent, and accordingly there may be a need for increased vigilance.

These recommendations are intended to complement national laws or regulations, accepted standards or particular procedures which may exist for specific trades, ships or types o f shipping operations.

It may not be practicable to apply all of these recommendations to all situations; howev er, when applying these recommendations becomes impracticable, every endeavour should be made to observe the intent of the recommendations, and attention should be paid to the h azards that may be involved in the specific entry and mitigations required to reduce the risks to an acceptable level.

1 INTRODUCTION The atmosphere in any enclosed space may be oxygen -deficient or oxygen-e nriched and/or contain flammable and/or toxic gases or vapours.

Such unsafe atmospheres could also subsequently occur in a space previously found to be safe.

Unsafe atmospheres are lik ely to --- URL: https://wwwcdn.imo.org/localresources/en/Know ledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf subsequently occur in a space previously found to be safe.

Unsafe atmospheres are lik ely to be present in spaces that are connected to a space con taining a hazardous atmosphere and may also be present in spaces adjacent to those space s where a hazard is known or suspected to be present.

1.1 An unsafe atmosphere can also quickly occur in a space previously found to be safe, including cargo holds and tanks where the hatch covers and tank covers have been open fo r some time.

Unsafe atmospheres can occur in frequently visited working areas and stores, such as for ecastle head spaces, pump-rooms, compressor rooms, inert gas rooms and other spaces that are adjacent to, or connected to, spaces containing a hazardous atmosphere.

These spaces include cargo holds containing cargo that depletes oxygen and /or emits tox ic, flammable or explosive gases, or is under fumigation.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING -- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/M SCResolutions/MSC.581(110).pdf flammable or explosive gases, or is under fumigation.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E NCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 3 I:\MSC\110\MSC 110-21-Add. 3.docx 1.2 An unsafe atmosphere can also occur in void spaces, cofferdams, pipe tunnels and also in accommodation areas and engine -rooms adjacent to or connected to cargo hold s containing hazardous cargo or under fumigation due to a failure in the means of sealin g the connection between the cargo space and the adjacent spaces.

1.3 The emission of toxic, flammable or explosive gases from cargo tanks, fuel tanks, sl op tanks and sewage tanks can also give rise to a potentially dangerous situation around open access hatches, manholes and ventilators on deck.

2 DEFINITIONS 2.1 Enclosed space means a space which may contain a hazardous atmosphere or lack of oxygen and has any of the following characteristics: --- URL: https://wwwcdn. imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(1 10).pdf 2.1 Enclosed space means a space which may contain a hazardous atmosphere or lac k of oxygen and has any of the following characteristics: .1 limited openings for entry and exit; .2 inadequate ventilation; or .3 not designed for continuous human occupancy.

This includes but is not limited to spaces that are diverse in their characteristics suc h as cargo holds, bilge spaces, ballast and other tanks, pump -rooms, chain lockers and engine crankcases.

2.2 Connected space means a space that is connected, by either permanent or temporary me ans (such as a door), to a source space that may contain a hazardous atmosphere.

For clarity, a space separated by a manual door, even if watertight, should be considere d as "connected" as it is impossible to tell from outside the space whether it is open o r closed or indeed properly sealed.

A connected space should be treated as containing a hazardous atmosphere until testing p roves otherwise.

The nature of the connection may lead to a --- URL: https://wwwcdn.imo.org/localresource s/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf atmosphere un til testing proves otherwise.

The nature of the connection may lead to a ■trapped hazardous atmosphere■.

2.3 Adjacent space means a space that shares a common boundary with a compartment that m
ay contain a hazardous atmosphere.

Such a space has no openings, temporary or permanent, into the hazardous compartment wha
tsoever and is designed to be a contiguous barrier.

Such a space may only contain a hazardous atmosphere in the event of failure of that bar
rier.

Precautions should relate to the possibility of such a failure.

2.4 Trapped Hazardous Atmosphere means a hazardous atmosphere that may be trapped in a c
onnected space in a manner that causes that space's atmosphere to fill and/or to empty a
t a different rate to the source space.

Such a space, while recognized as containing the same atmosphere, should be treated inde
pendently to the source space, and should be assumed to contain a hazardous atmosphere u
ntil proved otherwise by testing.

For example, --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIM
OResolutions/MSCResolutions/MSC.581(110).pdf assumed to contain a hazardous atmosphere u
ntil proved otherwise by testing.

For example, a trapped atmosphere may remain even after the cargo in the source space is
discharged.

2.5 Competent person means a person with an operational level of competency to make an i
nformed assessment of the likelihood of a dangerous atmosphere being present or subseque
ntly arising in the space.

2.6 Responsible person means a person in a management level on board a ship (i.e.

master, chief mate, chief engineer officer or second engineer officer) of competency and
authorized by the shipping company to permit entry into an enclosed space.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E
NCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 4 I:\MSC\110\MSC 110-21-Add.
3.docx 2.7 Attendant means a person maintains a watch over those entering the enclosed s
pace, to maintain communications with those inside the space and to initiate the emergen
cy --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutio
ns/MSCResolutions/MSC.581(110).pdf to maintain communications with those inside the spac
e and to initiate the emergency procedures in the event of an incident occurring.

2.8 Enclosed Space Register means a ship -specific register which lists all enclosed spa
ces on board the ship, along with their connected spaces and adjacent spaces, their haza
rds, associated risk mitigations if applicable, and how the atmosphere in these spaces m
ay change depending upon the nature of cargo carried or the content of the space, and wh
ich forms a part of the safety management for enclosed spaces.

3 SAFETY MANAGEMENT FOR ENTRY INTO ENCLOSED SPACES 3.1 A safety strategy should be adopt
ed in order to prevent accidents on entry into enclosed spaces in a comprehensive manner
by the company, in consultation with the ship.

This should include an Enclosed Space Register, which should be produced on a ship-by-sh
ip basis to identify enclosed spaces, the hazards of those spaces, assessment of risks u
nder the --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMORes
olutions/MSCResolutions/MSC.581(110).pdf basis to identify enclosed spaces, the hazards
of those spaces, assessment of risks under the differing conditions likely to arise in t
he space and the risk mitigation measures required to be put into place prior to entry.

This should also include an assessment of how the atmosphere in these enclosed spaces ma
y be impacted by the contents of the spaces themselves, such as in case a ship stores tr
eated sewage or grey water temporarily in its ballast water tanks, or the contents of co
nnected spaces or adjacent spaces, including cargo, fumigants, fuel oils, slops, oxygen-
depleting conditions and the physical or structural arrangement of the space.

3.2 The company should ensure that all relevant information relating to the hazards of t
he cargo, as submitted by the shipper in accordance with the applicable requirements of
SOLAS regulation VI/2, the International Maritime Solid Bulk Cargoes Code (IMSBC Code)
, the --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolu
tions/MSCResolutions/MSC.581(110).pdf regulation VI/2, the International Maritime Solid
Bulk Cargoes Code (IMSBC Code), the International Maritime Dangerous Goods Code (IMDG
Code), the IBC Code and the International Gas Carrier Code (IGC Code), is provided in
a format that is understandable to the ship's crew and distributed to those on board who

may be exposed to these hazards.

3.3 The company should ensure that the procedures for entering enclosed spaces are inclu
ded and implemented among the key shipboard operations concerning the safety of the pers
onnel and the ship, in accordance with paragraphs 6.4, 7 and 12.3 of the International S
afety Management (ISM) Code.

3.4 The company should ensure that all relevant crew members are given adequate training
in the safety management of enclosed spaces as per paragraphs 6.3 and 6.5 of the ISM Cod
e.

3.5 The company should ensure that adequate time has been allowed for any planned --- UR
L: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCRes
olutions/MSC.581(110).pdf ISM Code.

3.5 The company should ensure that adequate time has been allowed for any planned enclos
ed space activity, and that undue time pressure, either explicit or implied, is avoided
as this has been found to be a causal factor of many enclosed space accidents.

3.6 The company should elaborate a procedural implementation scheme which provides for c
omprehensive training in the maintenance, calibration and use of atmospheric testing equ
ipment in such spaces.

This training should be recorded, and individual crew member s who have been deemed comp
etent in the maintenance, calibration and use of atmospheric testing equipment should be
listed.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E
NCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 5 I:\MSC\110\MSC 110-21-Add.
3.docx 3.6.1 The competent and the responsible persons should be trained in enclosed spa
ce --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutio
ns/MSCResolutions/MSC.581(110).pdf Annex 28, page 5 I:\MSC\110\MSC 110-21-Add.3.docx 3.6
.1 The competent and the responsible persons should be trained in enclosed space hazard
recognition, evaluation, measurement, control and elimination, using standards acceptabl
e to the Administration.

The competent person should have received adequate training along with sufficient theore
tical knowledge and practical experience to make an informed assessment of the likelihoo
d of a dangerous atmosphere being present or subsequently arising in the space.

The responsible person should have sufficient knowle dge of the procedures to be establi
shed and complied with on board and received adequate training, in order to ensure that
an enclosed space is safe for entry and occupancy.

The attendant should be adequately trained within the safety management system.

3.6.2 All crew members should be trained, as appropriate, in enclosed space safety, ---
URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCR
esolutions/MSC.581(110).pdf 3.6.2 All crew members should be trained, as appropriate, in
enclosed space safety, including familiarization with onboard procedures for recognizing
, evaluating and controlling hazards associated with entry into enclosed spaces.

3.7 The company should identify and provide the necessary equipment such as ventilation
devices, atmosphere testing equipment, breathing apparatus and personnel recovery appara
tus to facilitate safe entry and rescue from an enclosed space.

The equipment provided should be suitable for the intended use.

A careful study of the operational profile should be carried out and all necessary equip
ment should always be on board in operational condition.

Consideration should be given to the appropriate use of technol ogy to assist in the haz
ard identification and mitigation of enclosed space entry.

Crew members should be trained in the use of equipment necessary to facilitate rescue fr
om an enclosed space and a record of this training should be maintained.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.581(110).pdf in the use of equipment necessary to facilitate rescue f
rom an enclosed space and a record of this training should be maintained.

3.8 The company should ensure that rescue drills from enclosed space identified in their
Enclosed Space Register are undertaken regularly as required by SOLAS regulation III/19.
3.6 using the equipment provided to facilitate a rescue from an enclosed space.

Such drills should focus on different aspects of operations involving enclosed spaces .

SOLAS regulation III/19.3.6 requires: .1 checking and use of personal protective equipme

nt required for entry; .2 checking and use of communication equipment and procedures; .3 checking and use of instruments for measuring the atmosphere in enclosed spaces; .4 chec king and use of rescue equipment and procedures; and .5 instructions in first aid and re suscitation techniques.

3.9 Internal audits by the company and external audits by the Administration of the ship 's --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutio ns/MSCResolutions/MSC.581(110).pdf 3.9 Internal audits by the company and external audit s by the Administration of the ship's safety management system should verify that the es tablished procedures 1 are followed and are consistent with the safety strategy referred to in these recommendations.

3.10 The company should establish criteria to manage additional risks involved during si multaneous operations (SIMOPS) where one of the operations includes an enclosed space en try.

The criteria should consider the assessment of personnel and resources availabl e in cas e of an emergency following an enclosed space entry.

1 Established procedures include resolution A.1050(27) as amended, the IMSBC Code, the B LU Code, the IMDG Code, the IBC Code, the IGC Code, Recommendations on the safe use of p esticides in ships applicable to the fumigation of cargo holds (MSC.1/Circ.1264), Guidel ines on tank entry for tankers using nitrogen as an inerting medium (MSC.1/Circ.1401) an d other accepted standards and industry codes of --- URL: https://wwwcdn.imo.org/localre sources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf nitroge n as an inerting medium (MSC.1/Circ.1401) and other accepted standards and industry code s of practice, as relevant.

 3.11 The company should ensure that single person entry into an enclosed space is not permitted.

4 IDENTIFICATION OF THE HAZARDS AND ASSESSMENT OF RISK 4.1 The company should ensure tha t a risk assessment is conducted to identify all enclosed spaces on board the ship and t hat the identified enclosed spaces are recorded in the Enclosed Space Register , which s hould be maintained on board the ship as well as ashore.

This Enclosed Space Register and risk assessment should be kept up to date as appropriat e to ensure its continued validity, particularly after loading and during the carriage o f cargoes --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMORe solutions/MSCResolutions/MSC.581(110).pdf to ensure its continued validity, particularly after loading and during the carriage of cargoes which may a dversely affect the safety of the atmosphere within a space.

A reassessment should also be made when the contents of the space change, such as in cas e a ship stores treated sewage or grey water temporarily in its ballast water tanks.

The company should consider the use of appropriate technology to assist in the hazard id entification and mitigation.

The Enclosed Space Register and risk assessment should form the basis of the development of the enclosed space emergency response plan (app endix 1).

It is recommended that the enclosed space emergency response plan be reviewed after each drill so that its effectiveness can be assessed and, if necessary, improvements made.

4.2 As entries for enclosed spaces other than cargo spaces may be different, there is a need to record information for the benefit of personnel involved.

Every ship should have an --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCen tre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf need to record information for the benefit of personnel involved.

Every ship should have an Enclosed Space Register, the information within which may form the basis of a r isk assessment.

The Enclosed Space Register should contain: .1 physical layout of the space and access a nd egress points, including of connected spaces, if any; .2 physical hazards in the spac e, e.g.

vertical ladders, unguarded openings, poor lighting, wet or slippery conditions, excessi ve heat; .3 connection to adjacent spaces; .4 specific hazards within the space, for exa mple, the effect of ballast water treatment method on the atmosphere within ballast tank s; .5 if used, information related to additional technology, helping to determine enclos

ed space condition; .6 information related to fixed and portable ventilation systems inc
luding equipment and where the equipment is stored; .7 estimated time taken to achieve t
he air changes for safe entry, using forced or natural ventilation; --- URL: https://www
cdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.5
81(110).pdf .7 estimated time taken to achieve the air changes for safe entry, using for
ced or natural ventilation; .8 lighting and means for temporary lighting including intri
nsically safe lighting where appropriate; .9 means for atmosphere testing; .10 any perti
nent information that would assist the risk assessment process; .11 locking and "Safe to
enter"/"Unsafe to enter" signage arrangements; and .12 the equipment necessary to facili
tate emergency rescue from the space.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E
NCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 7 I:\MSC\110\MSC 110-21-Add.
3.docx 4.3 In order to ensure safety from the outset, entry and occupancy, a competent p
erson should make an assessment of any potential hazards in the space to be entered, tak
ing into account the characteristics of the previous and current cargo carried with Safe
ty Data Sheets --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Indexof
IMOResolutions/MSCResolutions/MSC.581(110).pdf account the characteristics of the previo
us and current cargo carried with Safety Data Sheets (SDS) and cargo information, ventil
ation of the space, coating of the space and other relevant factors.

Opening of an enclosed space should be subject to a risk assessment, taking into conside
ration the potential for release of dangerous ga ses or potential to create an explosive
atmosphere: the competent person's assessment should determine the potential for the pre
sence of an oxygen -deficient, oxygen -enriched, flammable or toxic atmosphere, which in
cludes carbon monoxide (CO) and carbon dioxide (CO2) as well as other toxic or asphyxian
t gases.

The competent person should bear in mind that the ventilation procedures for an adjacent
or connected space may be different from those for the enclosed space itself.

The details of the assessment should be recorded in a standard format, and maintained on
board the ship.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.581(110).pdf details of the assessment should be recorded in a standa
rd format, and maintained on board the ship.

4.4 The nature of the hazards that may be present should be fully understood by those on
board the ship, both crew and shore-based personnel, and those ashore managing the ship.

4.5 The procedures to be followed for testing the atmosphere in the space and for entry
should be decided on the basis of the assessment.

These will depend on whether the assessment shows that: .1 there is minimal risk to the
health or life of personnel entering the space; or .2 there is no immediate risk to heal
th or life, but a risk could arise during the course of work in the space; or .3 a risk
to health or life is identified.

4.6 Where the assessment indicates minimal risk to health or life or potential for a ris
k to arise during the course of work in the space, the precautions described in sections
5 to 8 and 10 should be followed, as appropriate.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.581(110).pdf arise during the course of work in the space, the precau
tions described in sections 5 to 8 and 10 should be followed, as appropriate.

4.7 Where the assessment identifies a risk to life or health, if entry is to be made, th
e additional precautions specified in section 9 should also be followed.

4.8 Throughout the assessment process, there should be an assumption that the space to b
e entered is considered as hazardous until positively proved to be safe for entry.

5 AUTHORIZATION OF ENTRY 5.1 No person should open or enter an enclosed space unless aut
horized by the master or a responsible person nominated by the master and the appropriat
e safety procedures laid down for the particular ship have been followed.

5.2 Entry into enclosed spaces should be planned and the use of an entry permit system,
which may include the use of a checklist, should always be used where the assessment ---
URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCR
esolutions/MSC.581(110).pdf which may include the use of a checklist, should always be u
sed where the assessment identifies any risk to health or life.

Prior to the entry of a space, an enclosed space entry permit should be issued by the ma

ster or the responsible person nominated by the master, and complied with at all times b
y the competent person included in the work especially those who enter the space.

An example of the enclosed space entry permit is provided in appendix 2.

The validity of the permit should be specified based on the risk assessment and should n
ever be longer than eight hours.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E
NCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 8 I:\MSC\110\MSC 110-21-Add.
3.docx 5.3 Additional permits may be required depending upon the ship’s safety managemen
t system (SMS) and the activity being planned within the enclosed space.

These permits may --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Inde
xofIMOResolutions/MSCResolutions/MSC.581(110).pdf system (SMS) and the activity being pl
anned within the enclosed space.

These permits may include energy isolation permits, electrical isolation permit, hot wor
k permit, working at height permit or the completion of documents such as Job Hazard Ana
lysis.

5.4 Shore personnel should be provided with information on the hazards associated with e
nclosed spaces that may be present aboard the ship.

6 GENERAL PRECAUTIONS 6.1 Access to enclosed spaces should be carefully managed on ships
where shore personnel regularly enter cargo spaces to load and discharge cargoes, especi
ally on bulk carriers and general cargo ships.

These recommendations are complementary to the requirements of the International Ship an
d Port Facility Security (ISPS) Code.

It is recommended that a basic schematic plan of the ship’s spaces should be placed at t
he gangway or at other access points of the ship clearly indicating which spaces are saf
e to enter, --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMO
Resolutions/MSCResolutions/MSC.581(110).pdf gangway or at other access points of the shi
p clearly indicating which spaces are safe to enter, and which spaces are not.

The plan should outline the enclosed space entry procedures that should be complied with
by both ship and shore personnel before any entry is permitted.

The use of simple safe to enter and unsafe to enter symbols is recommended.

The plan should be time-stamped and dated and amended as necessary by the responsible pe
rson.

This may also be beneficial on ships where large numbers of shore personnel are bei ng e
mployed on the ship, such as in dry dock or ship repair facilities.

An example of a ship■s schematic plan is contained in appendix 3.

6.2 Before any personnel are authorized to enter any cargo space containing any cargo, t
he competent person should carry out a risk assessment of the cargo; identification of t
he physical characteristics of the cargo space(s) concerned; and the operations to be ca
rried out, and the responsible person should: --- URL: https://wwwcdn.imo.org/localresou
rces/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf physical c
haracteristics of the cargo space(s) concerned; and the operations to be carried out, an
d the responsible person should: .1 where it is necessary for shore personnel to enter a
ny such spaces, conduct a pre-operational risk assessment prior to commencement of cargo
operations or arrival of shore personnel.

When hazards are identified a joint risk assessment with terminal representatives respon
sible for operations on board the ship, or with other appropriate shore personnel, shoul
d be conducted; and .2 identify and agree to the precautions required during entry and e
nclosed space entry permit arrangements to be used.

6.3 Entry doors or access hatches leading to enclosed spaces should at all times be secu
red against entry unless the spaces have been risk assessed, atmospherically tested as r
equired and declared safe for entry.

The ship may use a system of seals similar to t hose --- URL: https://wwwcdn.imo.org/loc
alresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf req
uired and declared safe for entry.

The ship may use a system of seals similar to t hose recommended in the ISPS Code.

Suitable portable signage, which is easily understandable by the vessel’s crew and also
by shore personnel engaged on the ship at this time and in the port, indicating the haza
rds should be posted on entry doors or access hatches leading to an enclosed space.

An example of such signage is contained within appendix 3.

These signs should be updated when the space becomes safe for entry or when a safe space becomes unsafe.

6.4 A door or hatch cover which is opened to provide natural ventilation of an enclosed space may, wrongly, be taken to be an indication of a safe atmosphere and therefore it i s recommended to station an attendant at the entrance or use a mechanical or physical ba rrier, such as a locked bar or chain positioned across the opening with an attached warn ing sign, to --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIM OResolutions/MSCResolutions/MSC.581(110).pdf such as a locked bar or chain positioned ac ross the opening with an attached warning sign, to prevent accidental entry.

It is recommended that operations such as these are included in the watch handover activ ity.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E NCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 9 I:\MSC\110\MSC 110-21-Add. 3.docx 6.5 The master or the responsible person should determine that it is safe to ente r an enclosed space by ensuring that: .1 potential hazards have been identified in the r isk assessment and as far as possible isolated or made safe; .2 the space has been thoro ughly ventilated by natural or mechanical means to remove any toxic or flammable gases i dentified in the hazard identification process gases; .3 the atmosphere of the space has been tested using certified and calibrated instruments to ascertain that the space conta ins 20.9% oxygen.

If the testing --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Indexof IMOResolutions/MSCResolutions/MSC.581(110).pdf instruments to ascertain that the space c ontains 20.9% oxygen.

If the testing indicates that the level of oxygen is less than 20.9%, or indicates the p resence of even low levels of flammable or toxic gases, then the space should be treated as one where the atmosphere is known or suspected to be unsafe, as described in section 9; .4 the space has been declared safe for entry and properly illuminated; .5 a suitable system of communication between all parties for use during entry has been agreed and tes ted, and the evacuation signals have been agreed upon; .6 personnel entering the space a re wearing personal gas detection equipment that has been properly calibrated and is cap able of monitoring the levels of oxygen, carbon monoxide and any other gases identified in the risk assessment; .7 an attendant has been nominated and properly instructed; .8 r escue and resuscitation equipment has been positioned , tested and ready --- URL: https: //wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/ MSC.581(110).pdf .7 an attendant has been nominated and properly instructed; .8 rescue a nd resuscitation equipment has been positioned , tested and ready for immediate use at t he entrance to the space and a rescue plan detailing the rescue arrangements has been ag reed.

Where ship or shore personnel are working in more than one cargo space at the same time, the rescue equipment should be positioned at a designated central location.

In the event of an emergency in any one space, all personnel working in other spaces sho uld be instructed to stop work immediately and exit the space; .9 personnel are provided with the appropriate personal protective equipment for the entry and subsequent tasks; a nd .10 the required permits have been issued, authorizing entry.

6.6 The precautions in sub -paragraphs .6 to .8 of paragraph 6.

5 above may not be applicable to every situation described in this section.

The responsible person authorizing entry --- URL: https://wwwcdn.imo.org/localresources/ en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf applicable to e very situation described in this section.

The responsible person authorizing entry should determine whether personal gas detectors , an attendant and the positioning of rescue equipment at the entrance to the space are necessary.

6.7 Only trained and authorized personnel should be assigned the duties of entering, fun ctioning as attendants or acting as members of rescue teams.

Ships' crews with rescue and first aid duties should be drilled periodically as required by SOLAS regulation I II/19.3.6 in rescue and first aid procedures.

Training should include as a minimum: RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING ENCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex

28, page 10 I:\MSC\110\MSC 110-21-Add.3.docx .1 identification of the hazards likely to be faced during entry into enclosed spaces and whilst within the space, in particular th e rapidity with which --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf spaces and whilst within the space , in particular the rapidity with which oxygen may be depleted in a space by corrosion o r biological means especially in higher ambient temperatu res.

The pace at which oxygen gets depleted increases exponentially when ventilation is stopp ed (fan or other equipment in case of forced ventilation, and closing a vent or damper i n case of natural ventilation) or other characteristics or properties of the contents or structure of the space.

While higher ambient temperatures can increase the rate of oxygen depletion, it should b e assumed that the rate of depletion will always be very rapid, regardless of the ambien t temperature; .2 an explanation of how the pace at which oxygen is depleted will increa se exponentially when ventilation is stopped (fan or other equipment in case of forced v entilation, and closing a vent or damper in case of natural ventilation), or when hatch covers are closed; --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf forced ventilation, and closing a ven t or damper in case of natural ventilation), or when hatch covers are closed; .3 identif ication and use of the various sources of information on the hazards associated with ind ividual solid bulk and liquid bulk cargoes, and the precautions to be adopted when enter ing spaces containing such cargoes, or their residues; .4 awareness of the fact that whe n a person or persons in an enclosed space shows signs of adverse health effects, that t hey should always assume that these effects are due to an oxygen -depleted or toxic atmo sphere in the space, and that they should not enter it themselves; .5 recognition of the signs of adverse health effects caused by exposure to hazards during entry; .6 knowledge and experience in the use of personal protective equipment required for entry; .7 rescue , first aid, Cardio Pulmonary Resuscitation (CPR) techniques and evacuation procedures; --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf required for entry; .7 rescue, first aid, Cardio Pulmona ry Resuscitation (CPR) techniques and evacuation procedures; .8 knowledge and experience in the use of communication equipment and procedures; .9 knowledge and experience in the use of instruments for measuring the atmosphere; .10 knowledge and experience in the use of rescue equipment and procedures; .11 knowledge of the IMO/WHO/ILO Medical First Aid G uide for Use in Accidents involving Dangerous Goods (MFAG), where appropriate; and .12 k nowledge of use of emergency and the first aid equipment for chemical tankers (IBC Code section 14.3) and for gas carriers (IGC Code section 14.3), where appropriate.

6.8 All equipment used in connection with entry should be in good working condition and inspected and tested prior to use.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E NCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 11 I:\MSC\110\MSC 110-21-Add .3.docx --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOReso lutions/MSCResolutions/MSC.581(110).pdf REVISED RECOMMENDATIONS FOR ENTERING ENCLOSED SP ACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 11 I:\MSC\110\MSC 110-21-Add.3.docx 6. 9 As far as practicable, enclosed space entry should be carried out during hours of dayl ight or normal working hours of the ship to ensure ready availability of reserve personn el in the event of an emergency.

7 TESTING THE ATMOSPHERE 7.1 The gas detection equipment should be appropriate for the c argo that the ship has carried and is carrying considering information provided in the s hipper's declaration, the Safety Data Sheets (SDS), the IMDG Code, the IMSBC Code, Inter national Bulk Chemical Code (IBC Code) and the IGC Code.

The gas detection equipment, including devices for testing CO 2, should be capable of op erating correctly even in oxygen-depleted atmosphere.

7.2 Appropriate testing of the atmosphere of a space should be carried out with properly calibrated equipment by persons trained in the use of the equipment.

The manufacturers' --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf calibrated equipment by persons train ed in the use of the equipment.

The manufacturers' instructions should be strictly followed.

Testing of the atmosphere of a space should be carried out before any person enters the space and at regular intervals thereafter until all work is completed.

Where appropriate, the testing of the atmosphere of a space should be carried out at as

many different levels and areas as is necessar y to account for gas stratification, and obtain a representative sample of the atmosphere in the space.

In some cases, it may be difficult to test the atmosphere throughout the enclosed space without entering the space (e.g.

the bottom landing of a stairway or complex areas of the structure withi n the space) an d this should be taken into account.

7.3 All ships should carry at least two sets of gas detection equipment as required by S OLAS regulation XI -1/7, taking into account the Guidelines to facilitate the selection of --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutio ns/MSCResolutions/MSC.581(110).pdf SOLAS regulation XI -1/7, taking into account the Gui delines to facilitate the selection of portable atmosphere testing instruments for enclo sed spaces as required by SOLAS regulation XI-1/7 (MSC.1/Circ.1477).

Any ship which may carry a cargo capable of generating hazardous vapours and which requi res regular entry into the cargo space for cleaning or inspection should carry two sets of gas detection equipment in addition to those required by SOLAS regulation XI-1/7 for assessing the risk to personnel entering the space.

The use of flexible hoses or fixed sampling lines, which reach remote areas within the e nclosed space, may allow for safe testing without the need to enter the space.

7.4 After completion of a suitable risk assessment targeted to the space to be entered s teady readings of all of the following should be obtained: .1 20.9% oxygen by volume; .2 the level of carbon dioxide has been checked and is less than 0.5% by volume (5,000 ppm) ; --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolution s/MSCResolutions/MSC.581(110).pdf .1 20.9% oxygen by volume; .2 the level of carbon diox ide has been checked and is less than 0.5% by volume (5,000 ppm); Note: National require ments may differ when determining the safe atmosphere range for gases stated above.

.3 less than 1% of lower flammable limit (LFL) on a suitably sensitive combustible gas i ndicator, where the assessment has determined that there is potential for flammable gase s or vapours; and .4 less than 50% of the occupational exposure limit (OEL) of any toxic vapours and gases.

7.5 If these conditions cannot be met, additional ventilation should be applied to the s pace and retesting should be conducted after a suitable interval.

Entry should be allowed only after all the above conditions are met.

When the atmosphere remains, or is suspected to be unsafe, then the guidance contained w ithin section 9 should apply.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E NCLOSED SPACES ABOARD SHIPS --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeC entre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf RESOLUTION MSC.581(110) (ado pted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING ENCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 12 I:\MSC\110\MSC 110-21-Add.3.docx 7.6 Any gas testing should be carried out with ventilation to the enclosed space stopped, and after conditio ns have stabilized, in order to obtain accurate readings.

7.7 Where the assessment has determined that there is potential for the presence of toxi c gases and vapours, appropriate testing should be carried out, using fixed or portable gas or vapour detection and/or monitoring equipment.

The readings obtained by this equipment should be below the occupational exposure limits for the toxic gases or vapours given in accepted national or international standards, in accordance with paragraph 7.4.

It should be noted that testing for flammability or oxygen content does no t provide a s uitable means of measuring for toxicity, nor vice versa.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.581(110).pdf noted that testing for flammability or oxygen content do es no t provide a suitable means of measuring for toxicity, nor vice versa.

7.8 It should be emphasized that the internal structure of the space, cargo, cargo resid ues and tank coatings may also present situations where oxygen -deficient areas may exis t, and should always be suspected, even when an enclosed space has been satisfactori ly tested as being suitable for entry.

This is particularly the case for spaces where the path of the supply and outlet ventila tion is obstructed by structural members or cargo.

8 PRECAUTIONS DURING ENTRY 8.1 The atmosphere should be tested frequently whilst the spa ce is occupied, and persons should be instructed to leave the space should there be any sign of deterioration in the conditions.

8.2 Persons entering enclosed spaces should be provided with calibrated and tested perso nal gas detection instrument or instruments that monitor the levels of oxygen, carbon -- - URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MS CResolutions/MSC.581(110).pdf personal gas detection instrument or instruments that moni tor the levels of oxygen, carbon dioxide, flammable gases or vapours, toxic gases (inclu ding carbon monoxide), and any other gases identified in the risk assessment.

8.3 Ventilation should be maintained while the space is occupied.

After a break and before re-entry, the atmosphere must be retested, the results verified as being acceptable then recorded.

In the event of failure of the ventilation system, any persons in the space should leave immediately.

8.4 Particular care should be taken when working on piping and valves within the space w hich may contain hazardous gases, vapours or liquids.

If conditions change during the work, increased frequency of testing of the atmosphere s hould be performed.

Examples of changing conditions are as follows – increase or decrease in ambient tempera ture, the need to use oxygen fuel torches or other welding equipment, the need to use a mobile plant, other activities --- URL: https://wwwcdn.imo.org/localresources/en/Knowled geCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf oxygen fuel torches or ot her welding equipment, the need to use a mobile plant, other activities in the space tha t may involve development of vap our suc h as cleaning debris, collecting sediments, pai nting within the enclosed spaces, changes in the ship's trim and list while working with in the enclosed space.

Conditions can also deteriorate during work breaks because of changes to the ventilation , the dist urbance of bilge contents, sludge or stagnant water, or the leakage of extern al contaminants into the space.

Depending on the result of the testing, a decision should be made as to whether it is sa fe to continue working.

8.5 In the event of an emergency, under no circumstances should the attendant enter the space before help has arrived and the situation has been evaluated to ensure the safety of those entering the space to undertake rescue operations.

Only properly trained and equipped personnel should perform rescue operations in enclose d spaces.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.581(110).pdf personnel should perform rescue operations in enclosed s paces.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E NCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 13 I:\MSC\110\MSC 110-21-Add .3.docx 9 ADDITIONAL PRECAUTIONS FOR ENTRY INTO A SPACE WHERE THE ATMOSPHERE IS KNOWN OR SUSPECTED TO BE UNSAFE 9.1 In preparing to enter an enclosed space, every effort should be made to ensure that it is safe to do so in order to undertake the activity required w ithin the space.

Entry into enclosed spaces where the atmosphere is known or suspected to be unsafe requi res very careful consideration, including an assessment of the hazards, residual risks a nd mitigations that need to be undertaken.

Spaces that have not been tested should be considered unsafe for persons to enter.

9.2 When considering the planned work activity within an enclosed space where the --- UR L: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCRes olutions/MSC.581(110).pdf 9.2 When considering the planned work activity within an enclo sed space where the atmosphere is known or suspected to be unsafe, suitably designed and constructed breathing apparatus of positive pressure type should always be worn, and onl y personnel trained i n its use should be allowed to enter the space.

Air -purifying respirators, dust masks and canister face masks do not provide a supply o f clean air from a source independent of the atmosphere within the space and should not be used.

Emergency Escape Breathing Devices (EEBDs) are not suitable to use for entry into enclos ed spaces.

9.3 Notwithstanding paragraph 6.5.6, persons entering enclosed spaces that may contain a suspected atmospheric hazard should be provided with calibrated and tested personal gas detector suitable for the gas or gases assessed as likely to occur in the space.

9.4 Rescue harnesses should be worn and unless impractical, lifelines should also be --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCR esolutions/MSC.581(110).pdf 9.4 Rescue harnesses should be worn and unless impractical, lifelines should also be used.

A means to facilitate evacuation from the enclosed space should be available and ready f or use, as per emergency response plan.

9.5 Appropriate protective clothing should be worn, particularly where there is any risk of toxic substances or chemicals coming into contact with the skin of those entering the space.

9.6 The advice in section 11 concerning actions to be taken in an emergency is particula rly relevant in this context.

10 HAZARDS RELATED TO SPECIFIC TYPES OF SHIPS OR CARGO 10.1 Dangerous goods in packaged form 10.1.1 The atmosphere of any space containing dangerous goods may put at risk the h ealth or life of any person making an entry.

Dangers may include flammable, toxic, corrosive or asphyxiant gases or vapours, residues on packages and spilled material.

The same haz ards may be present in spaces adjacent or connected to the cargo spaces.

Information on the --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Ind exofIMOResolutions/MSCResolutions/MSC.581(110).pdf may be present in spaces adjacent or connected to the cargo spaces.

Information on the hazards of specific substances is contained in the IMDG Code, the SDS and the Shipper’s Declaration.

If there is evidence or suspicion that leakage of dangerous sub stances has occurred, th e precautions specified in these recommendations should be followed.

10.1.2 Personnel required to deal with spillages or to remove defective or damaged packa ges should be appropriately trained and wear suitable breathing apparatus and protective clothing commensurate with the task.

10.2 Liquid bulk The industry has produced extensive advice to managers, operators and c rews of ships engaged in the bulk carriage of oil, chemicals and liquefied gases, in the form of specialist safety guides.

Information contained in the guides on enclosed space entry am plifies these recommendat ions.

In particular, for chemical tankers as defined in SOLAS regulation VII/8, the --- URL: h ttps://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolut ions/MSC.581(110).pdf recommendations.

In particular, for chemical tankers as defined in SOLAS regulation VII/8, the diversity of bulk liquid chemicals carried in some cargo tanks and the limitations in vapour detec tion technologies may pose complications that require specific and targeted mitigation.

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E NCLOSED SPACES ABOARD SHIPS MSC 110/21/Add.3 Annex 28, page 14 I:\MSC\110\MSC 110-21-Add .3.docx 10.2.1 Safety information Safety information2 for the correct handling and carri age of liquid bulk and gas cargoes is set out in the SDS or other cargo information prov ided by the shipper to the master.

Such information should be made available to all ship and shore personnel involved in th e handling of the cargo, cargo slops and tank cleaning.

10.2.2 Use of nitrogen as an inert gas3 Nitrogen is a colourless and odourless gas that, when used as an inert gas, causes oxygen --- URL: https://wwwcdn.imo.org/localresources/ en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf 10.2.2 Use of n itrogen as an inert gas3 Nitrogen is a colourless and odourless gas that, when used as a n inert gas, causes oxygen deficiency in enclosed spaces and at exhaust openings on deck during purging of tanks and void spaces and use in cargo holds.

It should be noted that one deep breath of 100% nitrogen gas will prove fatal.

10.3 Solid bulk On ships carrying solid bulk cargoes, dangerous atmospheres may develop within cargo spaces, connected and adjacent spaces.

The hazards may include flammability, toxicity, oxygen depletion, carbon dioxide and/or carbon monoxide generation, or self -heating, as identified in the shipper's declaration and/or in the individual schedules in appendix 1 of the IMSBC Code.

Solid bulk cargoes listed in the IMSBC Code should be carried in accordance with the pro visions of the Code, including precautions, atmosphere t esting, ventilation and other r equirements specified.

Solid bulk cargoes that are not listed in the IMSBC Code should --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf other requirements specified.

Solid bulk cargoes that are not listed in the IMSBC Code should be carried in accordance with section 1.3 of the Code, including the conditions for carriage and handling as dete rmined by the relevant co mpetent authorities.

Grain cargoes and timber cargoes other than those listed in the IMSBC Code 4 may also ca use oxygen depletion and toxic gas emissions, primarily carbon dioxide, in cargo holds a nd connected and adjacent spaces.

10.3.1 Enclosed hold access trunks 10.3.1.1 In certain designs of hold access, a stair a rrangement, sometimes referred to as the "Australian Ladder", substitutes for a vertical ladder.

On some ships these are contained within enclosed protective structures that are open on ly at the top and bottom o f the hold.

When a cargo that presents the risk of a hazardous atmosphere is loaded, this "connected " space will quickly adopt atmospheric attributes similar to that of the source space.

The enclosed --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.581(110).pdf quickly adopt atmospheric attributes simila r to that of the source space.

The enclosed Australian ladder access should not be used unti l it has been fully ventil ated, tested and confirmed to be safe.

It should be noted that such spaces are difficult to ventilate, unless cargo blocking th e bottom of the ladder is removed.

10.3.1.2 When hatches are opened to ventilate the above cargo space at discharge, the ha zardous atmosphere often gets trapped in the access trunk.

2 16.2.3.1 of the IBC Code and 18.3.1 of the IGC Code.

3 Refer to the Guidelines on tank entry for tankers using nitrogen as an inerting medium (MSC.1/Circ.1401).

4 Refer to the International Code for the Safe Carriage of Grain in Bulk (resolution MSC .23(59) (amended by resolution MSC.552(108)) and the Code of Safe Practice for Ships Car rying Timber Deck Cargoes, 2011 (resolution A.1048(27)).

RESOLUTION MSC.581(110) (adopted on 27 June 2025) REVISED RECOMMENDATIONS FOR ENTERING E NCLOSED SPACES ABOARD SHIPS --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeC entre/IndexofIMOResolutions/MSCResolutions/MSC.578(110).pdf RESOLUTION MSC.578(110) (ado pted on 27 June 2025) ENCOURAGEMENT OF MARITIME INFORMATION-SHARING THROUGH THE USE OF N ATIONAL AND REGIONAL MARITIME INFORMATION-SHARING CENTRES TO ENHANCE MARITIME SAFETY AND SECURITY MSC 110/21/Add.1 Annex 9, page 1 I:\MSC\110\MSC 110-21-Add.1.docx ANNEX 9 RESOL UTION MSC.578(110) (adopted on 27 June 2025) ENCOURAGEMENT OF MARITIME INFORMATION-SHARI NG THROUGH THE USE OF NATIONAL AND REGIONAL MARITIME INFORMATION-SHARING CENTRES TO ENHA NCE MARITIME SAFETY AND SECURITY THE MARITIME SAFETY COMMITTEE, RECALLING article 1(a) o f the Convention on the International Maritime Organization, according to which the purp ose of the Organization is to provide machinery for cooperation among Governments in the field